

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO5:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL5)

Assessment Tool Weightage: 40%

QUESTIONS: 3

Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

*I M.HARINI(192424011) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

M.HARINI

Annexure A

Constraint-Based Problem Solving

1. A multinational company is opening a new branch office in a remote rural location where high-speed wired internet is unavailable. The office must support cloud-based applications, video conferencing, and secure communication with the headquarters. The company has a limited budget for connectivity but requires uninterrupted business operations. As a network engineer, recommend a suitable internet connection, networking devices, and security mechanisms. Justify your recommendations by considering bandwidth limitations, reliability, cost, and future scalability.
2. A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, and hostels. The network must support a large number of simultaneous users while minimizing signal interference and ensuring secure access. Due to budget constraints, only a limited number of wireless access points can be deployed. Analyze the requirements and propose an efficient wireless network design, including access point placement, frequency band selection, and authentication methods. Explain how your solution balances coverage, performance, security, and cost.
3. A financial institution is upgrading its internal computer network to improve cybersecurity while allowing employees to access sensitive customer information and online banking services. The organization must comply with strict security policies without significantly affecting network performance or increasing operational costs. Design a secure network architecture by selecting appropriate network segmentation techniques, firewall deployment, authentication methods, and intrusion detection mechanisms. Justify your design by considering the constraints of security, performance, compliance, and budget.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Problem & Constraints	20%	Clearly identifies all constraints and fully understands the problem requirements	Identifies most constraints with minor gaps in understanding	Identifies some constraints but misses key aspects	Shows little or no understanding of constraints or problem
Problem Decomposition	15%	Breaks problem into logical, well-structured sub-problems effectively	Breaks problem into sub-parts with some logical flow	Attempts decomposition but lacks clarity or structure	No clear decomposition of the problem
Application of Constraints in Solution	20%	Applies all constraints correctly and consistently in solution design	Applies most constraints correctly with minor errors	Applies some constraints but with noticeable errors	Fails to apply constraints appropriately
Logical Reasoning & Strategy	15%	Uses strong, clear, and efficient reasoning strategies	Uses generally sound reasoning with minor gaps	Reasoning is partially correct but inconsistent	Reasoning is unclear or incorrect
Solution Accuracy & Feasibility	15%	Solution is fully correct, feasible, and optimized	Solution is mostly correct and feasible	Solution has errors or is partially feasible	Solution is incorrect or not feasible
Creativity & Innovation	5%	Demonstrates highly innovative and unique approach	Shows some creativity in approach	Limited creativity shown	No creativity; relies on basic or copied ideas
Clarity of Presentation	10%	Solution is clearly presented with excellent organization and explanation	Presentation is clear with minor issues	Presentation lacks clarity or organization	Poorly presented and difficult to understand

1. Rural Branch Office – Internet Connectivity and Security

Problem Analysis

The multinational company is opening a branch office in a remote rural area where high-speed wired Internet such as fiber is not available. The office must support cloud applications, video conferencing, file sharing, email, and secure communication with headquarters. At the same time, the company has a limited budget and requires reliable connectivity.

Recommended Internet Connection

The best solution is 4G/5G Fixed Wireless Access (FWA) as the primary Internet connection.

- A 4G/5G router can provide Internet connectivity without requiring fiber or cable installation.
- 5G should be preferred if good coverage is available because it provides higher bandwidth and lower latency.
- If 5G is unavailable, 4G LTE can be used.
- A satellite Internet connection can be added as a backup where cellular coverage is unreliable.
- Automatic failover can switch to the backup connection when the primary connection fails.

Recommended Networking Devices

1. 4G/5G Router

- Connects the office to the Internet.
- Supports SIM-based cellular connectivity.
- Provides routing and basic security.

2. Next-Generation Firewall

- Protects the internal network from external attacks.
- Controls incoming and outgoing traffic.
- Can provide VPN, intrusion prevention, and application filtering.

3. Managed Network Switch

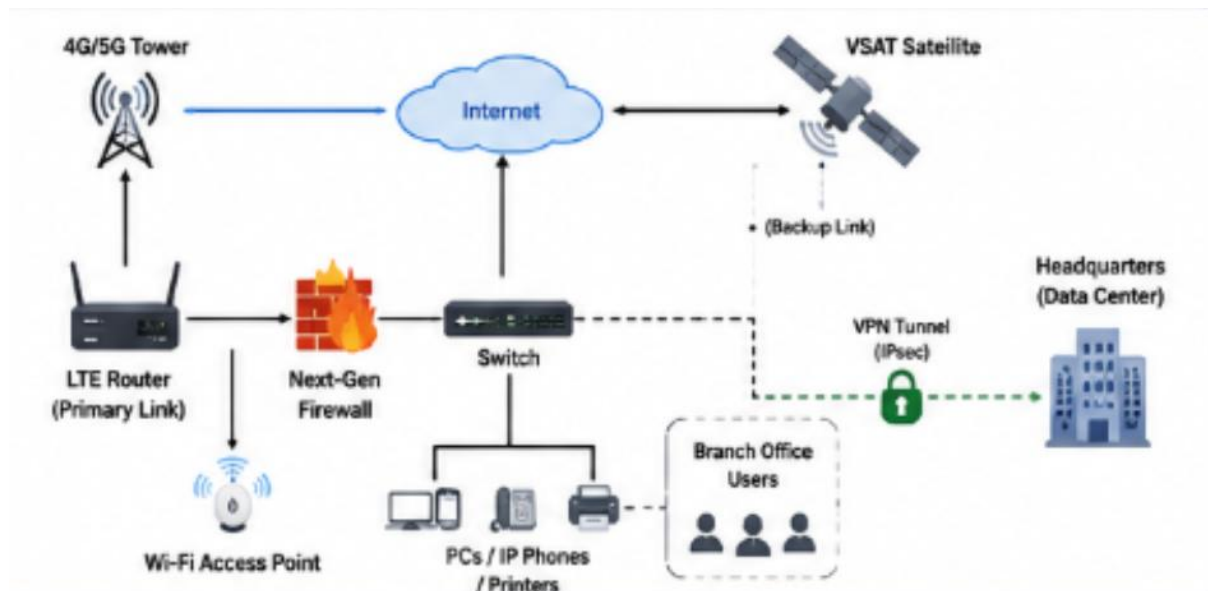
- Connects computers, printers, IP phones, and other wired devices.
- Supports VLANs for separating different types of users.

4. Wi-Fi 6 Access Point

- Provides high-speed wireless connectivity.
- Supports multiple employees simultaneously.

5. UPS

- Provides temporary power during power failures.
- Prevents sudden shutdown of routers, switches, and firewalls.



Security Mechanisms

- Site-to-site VPN should be established between the branch office and headquarters.
- Firewall should block unauthorized traffic.
- WPA3 should be used for wireless security.
- Strong passwords and multi-factor authentication (MFA) should be implemented.
- VLANs should separate employee, guest, and administrative traffic.
- Antivirus and endpoint security should be installed on computers.
- Network activity should be monitored regularly.

Constraint-Based Justification

Constraint	Solution
Limited budget	Use 4G/5G instead of expensive wired infrastructure
Rural location	Fixed wireless or satellite connectivity
Reliability	4G/5G + satellite backup
Security	Firewall + VPN + WPA3 + MFA
Video conferencing	5G provides suitable bandwidth and low latency
Cloud applications	Reliable Internet with traffic prioritization
Future scalability	Upgrade to better 5G plans or fiber when available

Conclusion

A 4G/5G-based network with a backup satellite connection, firewall, managed switch, Wi-Fi 6

AP, and VPN provides a cost-effective and secure solution. It meets the current requirements while allowing the company to upgrade the network in the future.

2. University Campus – Large-Scale Wireless Network

Problem Analysis

The university needs to provide wireless Internet access to more than 3,000 students across classrooms, laboratories, libraries, and hostels. The major challenges are:

- Large number of simultaneous users
- Limited number of access points
- Signal interference
- High bandwidth requirements
- Secure authentication
- Limited budget

Therefore, the university needs a high-density wireless network design rather than simply installing APs everywhere.

Recommended Wireless Technology

The university should use Wi-Fi 6 or Wi-Fi 6E access points.

Wi-Fi 6 is suitable because it provides:

- Higher capacity
- Better performance with many users
- Lower latency
- Improved efficiency
- Better handling of crowded networks

Wi-Fi 6E can additionally use the 6 GHz spectrum where supported.

Access Point Placement

Access points should be placed based on user density rather than only geographical coverage.

Classrooms

- Install APs near classroom areas.
- High-density classrooms should receive priority.
- Avoid placing too many APs on the same channel.

Laboratories

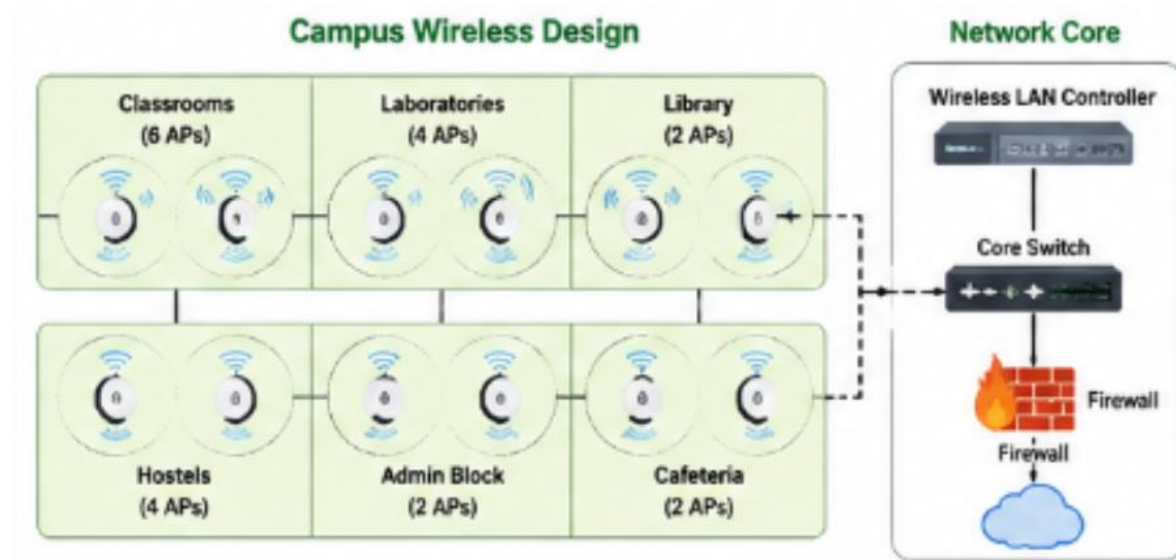
- Laboratories require high bandwidth because many computers may be connected simultaneously.
- APs should be positioned to provide stable connectivity to all devices.

Library

- Libraries generally have many simultaneous users.
- High-capacity APs should be installed in reading areas.

Hostels

- Hostels require wireless coverage across rooms and common areas.
- APs should be distributed to avoid excessive load on individual APs.



Frequency Band Selection

2.4 GHz

Advantages:

- Longer range
- Better wall penetration

Disadvantages:

- Limited number of channels
- More interference
- Lower capacity

Therefore, 2.4 GHz should mainly be used for devices that require longer range.

5 GHz

- Higher speed
- More available channels
- Less interference than 2.4 GHz
- Suitable for classrooms, laboratories, and libraries

6 GHz

- Provides additional spectrum.
- Suitable for modern Wi-Fi 6E devices.
- Useful for high-density environments.
- Can reduce congestion on 2.4 GHz and 5 GHz.

Interference Management

To minimize interference:

- Use proper channel planning.
- Avoid assigning the same channel to nearby APs.
- Reduce unnecessary transmission power.
- Use automatic channel management where appropriate.
- Monitor AP utilization.
- Prefer 5 GHz and 6 GHz for high-bandwidth users.

Authentication and Security

The university should use **WPA3-Enterprise with 802.1X and RADIUS**.

Students can authenticate using their university credentials.

Separate networks should be created:

- **Student VLAN**
- **Faculty VLAN**
- **Administrative VLAN**
- **Guest VLAN**
- **Laboratory/IoT VLAN**

This prevents one group of users from directly accessing another group's resources.

Centralized Management

A wireless controller or cloud-managed Wi-Fi system can be used to:

- Monitor APs
- Detect interference
- Manage channels
- Monitor connected users
- Identify overloaded APs
- Apply security policies
- Troubleshoot network problems

Constraint-Based Justification

Constraint	Solution
More than 3,000 students	High-density Wi-Fi 6/6E
Limited APs	Place APs according to user density
Signal interference	Channel planning and power control
High bandwidth	Prefer 5 GHz/6 GHz
Security	WPA3-Enterprise + 802.1X
Different user groups	VLAN segmentation

Constraint	Solution
Limited budget	Optimize AP placement instead of excessive deployment
Future growth	Wi-Fi 6E and centralized management

Conclusion

A **Wi-Fi 6/6E high-density wireless network** with strategic AP placement, 5/6 GHz usage, proper channel planning, WPA3-Enterprise authentication, and VLAN segmentation provides a good balance between **coverage, performance, security, and cost**.

3. Financial Institution – Secure Network Architecture

Problem Analysis

A financial institution handles highly sensitive information such as:

- Customer personal information
- Bank account information
- Transaction data
- Employee credentials
- Online banking services

Therefore, the network must provide strong cybersecurity, controlled access, high performance, compliance, and reasonable operating costs.

Proposed Network Architecture

The network should be divided into multiple security zones.

1. Employee Network

Employees' computers should be placed in a separate VLAN.

They should have access only to the applications and databases required for their jobs.

2. Customer/Application Network

Customer-facing applications and banking services should be isolated from normal employee systems.

3. Database Network

Sensitive customer databases should be placed in a highly protected network segment.

Only authorized application servers should be allowed to communicate with the database.

4. Guest Network

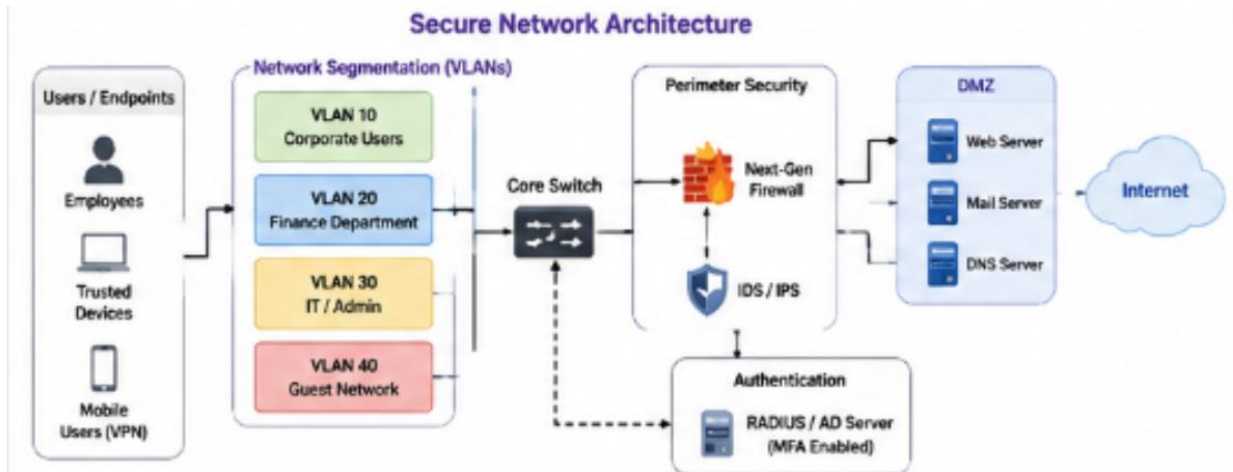
Guest devices should have Internet-only access.

They should not be able to access internal systems.

5. Management Network

Network devices such as switches, routers, firewalls, and servers should be managed through a separate secure network.

Network Segmentation



Firewall Deployment

A Next-Generation Firewall (NGFW) should be placed at the Internet boundary.

It should provide:

- Packet filtering
- Application control
- Malware protection
- Intrusion prevention
- VPN support
- URL filtering
- Traffic monitoring

Additional firewall rules should be applied between highly sensitive internal networks.

For example:

Employee VLAN → Application Server: Allowed only for required applications.

Employee VLAN → Database: Direct access should generally be blocked.

Guest VLAN → Internal Network: Completely blocked.

Internet → Database: Blocked.

Authentication Mechanisms

Multi-Factor Authentication

MFA should be required for sensitive systems.

Users can authenticate using:

1. Username and password
2. OTP/security key/authenticator

This provides additional protection even if a password is stolen.

802.1X Authentication

802.1X can be used for controlling access to wired and wireless networks.

Only authenticated employees and authorized devices can connect to internal networks.

Role-Based Access Control

Employees should receive permissions according to their roles.

For example:

- Accountant → Financial systems
- Customer service → Customer support systems
- IT administrator → Network management
- Manager → Approved business applications

Employees should not receive unnecessary permissions.

Intrusion Detection and Prevention

The institution should deploy **IDS/IPS**.

IDS – Intrusion Detection System

IDS monitors network traffic and identifies suspicious activities.

Examples:

- Port scanning
- Unusual traffic
- Malware communication
- Unauthorized access attempts

IPS – Intrusion Prevention System

IPS can automatically block suspicious traffic.

For example:

Attacker

|

↓

Internet

|

Firewall

|

IDS/IPS → Detects suspicious traffic

|

X → Blocks attack

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Internal Network

Logging and Monitoring

Centralized monitoring should be used to collect:

- Firewall logs
- Authentication logs
- Server logs
- IDS/IPS alerts
- Failed login attempts
- Network traffic information

A **SIEM system** can analyze these logs and generate alerts for suspicious activities.

Constraint-Based Justification

Constraint	Recommended Solution
High security	Firewall + IDS/IPS + segmentation
Sensitive customer data	Separate database VLAN/security zone
Unauthorized access	MFA + 802.1X + RBAC
Compliance	Centralized logging and access monitoring
Network performance	VLANs and controlled traffic policies
Limited budget	Use logical segmentation instead of completely separate physical networks
Attack prevention	NGFW + IPS
Future scalability	Expand VLANs, firewall rules, and security zones

Conclusion

The financial institution should implement a segmented network architecture using VLANs, security zones, NGFW, MFA, 802.1X, IDS/IPS, and centralized monitoring. This design protects sensitive customer information while maintaining network performance and controlling operational costs. It also provides a scalable architecture that can support future growth and stricter security requirements.